

Mean and Variance of Binomial Distribution $X \sim B(n, p)$

$$\begin{aligned}
 E[X] &= \sum_{x=0}^n x \binom{n}{x} p^x (1-p)^{n-x} \\
 &= \sum_{x=1}^n x \frac{n!}{x!(n-x)!} p^x (1-p)^{n-x} \\
 &= \sum_{x=1}^n \frac{x}{x} \frac{n!}{(x-1)!(n-x)!} p^x (1-p)^{n-x}
 \end{aligned}$$

$x=0$ term removed

let $y = x-1$ and $m = n-1$

$$\begin{aligned}
 E[X] &= \sum_{y=0}^m \frac{(m+1)!}{y!(m-y)!} p^{y+1} (1-p)^{m-y} \\
 &= (m+1) p \sum_{y=0}^m \frac{m!}{y!(m-y)!} p^y (1-p)^{m-y} \\
 &= n p \sum_{y=0}^m \binom{m}{y} p^y (1-p)^{m-y} \\
 &= n p (p + (1-p))^m = 1
 \end{aligned}$$

Binomial expansion

$$\therefore E[X] = np$$

Similarly, we use $E[X^2] - E[X]^2$ to find σ_x^2

$$\begin{aligned}
 \text{Var}[X] &= E[X^2] - E[X]^2 \\
 &= E[X(X-1)] + E[X] - E[X]^2
 \end{aligned}$$

use the same approach as above, but using $y = x-2$ and $m = n-2$, we get $E[X(X-1)] = n(n-1)p^2$

$$\begin{aligned}
 \therefore \text{Var}[X] &= n(n-1)p^2 + np - n^2p^2 \\
 &= np(1-p)
 \end{aligned}$$

Poisson Distribution - Approximation to Binomial Distribution $X \sim B(n, p)$, $\mu = np$

For large n :

$$\lim_{n \rightarrow \infty} \binom{n}{x} p^x (1-p)^{n-x}$$

and let $p = \frac{\mu}{n}$

$$= \lim_{n \rightarrow \infty} \frac{n!}{(n-x)! x!} \left(\frac{\mu}{n}\right)^x \left(1 - \frac{\mu}{n}\right)^{n-x}$$

$$= \lim_{n \rightarrow \infty} \underbrace{\frac{n!}{(n-x)!}}_{\substack{\rightarrow 1 \text{ as } n \rightarrow \infty \\ \text{see below}}} \underbrace{\frac{1}{n^x}}_{\substack{\rightarrow 0 \text{ as } n \rightarrow \infty \\ \text{see below}}} \underbrace{\frac{\mu^x}{x!}}_{\text{constant}} \underbrace{\left(1 - \frac{\mu}{n}\right)^n}_{\substack{\rightarrow e^{-\mu} \text{ as } n \rightarrow \infty \\ \text{see below}}} \underbrace{\left(1 - \frac{\mu}{n}\right)^{-x}}_{\rightarrow 1 \text{ as } n \rightarrow \infty}$$

$$\frac{n(n-1)(n-2)\dots(n-x+1)}{n \cdot n \cdot n \dots n \text{ (x terms)}}$$

$\rightarrow 1 \text{ as } n \rightarrow \infty$

$$\lim_{n \rightarrow \infty} \left(1 - \frac{\mu}{n}\right)^n = e^{-\mu}$$

$$= \frac{\mu^x}{x!} e^{-\mu} \rightarrow \text{Poisson distribution where } \mu = np$$

Mean and Variance of Geometric Distribution $X \sim G(p)$

$$E[X] = \sum_{x=1}^{\infty} x (1-p)^{x-1} p$$

$$\text{let } q = 1-p$$

$$\therefore E[X] = p \sum_{x=1}^{\infty} x q^{x-1} \rightarrow \frac{dq^x}{dq} = x q^{x-1}$$

$$= p \frac{d}{dq} \underbrace{\sum_{x=0}^{\infty} q^x}_{\text{infinite geometric series}} = \frac{1}{1-q}$$

$$= p \frac{d}{dq} (1-q)^{-1} = \frac{p}{(1-q)^2}$$

$$= \frac{p}{p^2} = \frac{1}{p}$$

$$\text{Var}[X] = E[X^2] - E[X]^2 = E[X(X-1)] + E[X] - E[X]^2$$

Similar approach to what we did for finding $\text{Var}[X]$ for Binomial Dist.

$$E[X(X-1)] = \sum_{x=1}^{\infty} x(x-1) p (1-p)^{x-1}$$

Similar to the above approach for $E[X]$

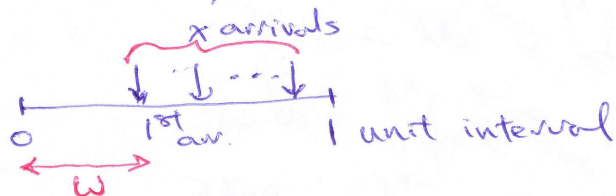
$$\text{we will get } \frac{2(1-p)}{p^2}$$

$$\therefore \text{Var}[X] = \frac{2(1-p)}{p^2} + \frac{1}{p} - \left(\frac{1}{p}\right)^2 = \frac{1-p}{p^2}$$

If the number of arrivals in an interval is Poisson distributed, show that the interarrival times are exponentially distributed.

let X = no. of arrivals in 1 unit interval

$$X \sim \text{Pois}(\mu), \text{ i.e. } P(X=x) = \frac{e^{-\mu} \mu^x}{x!}$$



w is the 1st arrival time from $t=0$

Q.V. W = interarrival time and we want to find the p.d.f. of W

$$\text{CDF } P(W \leq w) = 1 - P(W > w)$$

$$= 1 - P(\text{no arrival in } (0, w])$$

$$= 1 - P(X=0 \text{ in } (0, w])$$

Poisson with mean $= \mu w$

$$= 1 - e^{-\mu w}$$

$$\therefore \text{pdf} = \frac{d}{dw} \text{CDF} = \boxed{\mu e^{-\mu w}}$$

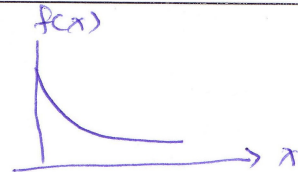
exponential dist

where μ = mean arrival rate

OR $\frac{1}{\mu}$ = mean interarrival time

Mean and Variance of Exponential Distribution $X \sim \exp(\lambda)$

$$f(x) = \lambda e^{-\lambda x}, \quad x > 0$$



$$E[X] = \int_0^{\infty} x \lambda e^{-\lambda x} dx$$

$$= \underbrace{-x e^{-\lambda x}}_0 \Big|_0^{\infty} + \underbrace{\int_0^{\infty} e^{-\lambda x} dx}$$

$$= 0 + \frac{e^{-\lambda x}}{-\lambda} \Big|_0^{\infty} = \frac{1}{\lambda}$$

(integration by part
 $u = x, \quad du = dx$
 $dv = \lambda e^{-\lambda x}$
 $v = -e^{-\lambda x}$)

$$\text{Var}[X] = \int_0^{\infty} (x - \frac{1}{\lambda})^2 \lambda e^{-\lambda x} dx$$

$$= -\underbrace{(x - \frac{1}{\lambda})^2 e^{-\lambda x}}_0 \Big|_0^{\infty} - \int_0^{\infty} -e^{-\lambda x} 2(x - \frac{1}{\lambda}) dx$$

$$= \frac{1}{\lambda^2} + \underbrace{\frac{2}{\lambda} \int_0^{\infty} x \lambda e^{-\lambda x} dx}_{E[X] = \frac{1}{\lambda}} - \underbrace{\frac{2}{\lambda} \int_0^{\infty} e^{-\lambda x} dx}_{= \frac{1}{\lambda}}$$

$$= \frac{1}{\lambda^2} +$$

$$= \frac{1}{\lambda^2}$$

$$\frac{2}{\lambda^2} - \frac{2}{\lambda^2} = 0$$

int by part
 $u = (x - \frac{1}{\lambda})^2$
 $du = 2(x - \frac{1}{\lambda}) dx$
 $dv = \lambda e^{-\lambda x} dx$
 $v = -e^{-\lambda x}$